

Roben Bhatti

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🇮🇹 Italian citizenship

Experience

German Aerospace Center (DLR)

Oct 2024 - Sep 2025

Data Scientist Intern

Bremen, Germany

- Developed and implemented Machine Learning models to quantify uncertainty in aerodynamic coefficients for reusable launch vehicles, leading to more robust estimation and performance analysis.
- Built a **CI/CD** pipeline using **Gitlab** and **Docker**, automating the deployment process and ensuring reliable releases.
- Collaborated within an **Agile team** to deliver data-driven results, culminating to the successful completion of a **co-authored** conference paper (IAC 2025) detailing the novel application.
- Addressed computational bottlenecks in model training by implementing Sparse Gaussian Processes, resulting in a **30% decrease in model training time** and enhancing the feasibility of large-scale uncertainty analysis.

Education

University of Padua

Oct 2022 - Jul 2025

Master of Science in Physics "Physics of Data" (GPA: 4.00 / 4.00)

Padua, Italy

- **Relevant Coursework:** Mathematical and Numerical Methods, Deep Learning and Neural Networks, Advanced Statistics for Physics Analysis, Information Theory and Inference, General Relativity.

University of Padua

Oct 2019 - Oct 2022

Bachelor of Science in Astronomy

Padua, Italy

- **Relevant Coursework:** Advanced Calculus, Statistics, Analytical Mechanics, Quantum Physics, Special Relativity.

Projects

LLM-Powered RAG Q&A System (Ongoing)

- Designed and implemented a **Retrieval-Augmented Generation (RAG)** pipeline using **LangChain** and **LangGraph**, enabling question-answering over uploaded documents (PDF, DOCX, TXT) and web content.
- Integrated Ollama for running local LLMs with **OpenAI API** fallback, returning answers with quoted citations from the original sources.
- Developed a **REST API** backend with **FastAPI**, following OOP design principles, and auto-documented endpoints via OpenAPI.

Streaming Particle Physics Data and Online Data Analysis with PySpark and Apache Kafka

- Engineered a distributed streaming pipeline capable of processing over **10 GB** of data per day, simulating a real-time detector stream with **Kafka** and **Spark** for live analysis.
- Deployed the environment on a cloud cluster using **AWS S3** for storage, Kafka for message queuing, and Spark for distributed processing.
- Developed a real-time interactive dashboard with Bokeh for **Data Visualization**.

Chess Position Recognition using Transformers (DETR) in PyTorch

- Developed a computer vision pipeline to digitize physical chess games, achieving **87.5%** accuracy in identifying pieces and their positions from a single image.
- Fine-tuned a DETR (DEtection TRansformer) model, leveraging **transfer learning** to significantly reduce training time and improve performance on custom-built dataset.
- Developed a robust **post-processing** module to automatically convert the model's raw detection output into standard notation (FEN), making the board state immediately compatible with chess engines and analysis tools.

Technical Skills

Languages: Python, R, SQL

Technologies: Docker, Git, CI/CD, Kafka, Spark, Pytorch, SciPy, NumPy, Pandas, LangChain, FastAPI, Tableau.

Languages

English: Professional, **Italian:** Native